

Knowing When Not to Answer: Noise-Conditional Hallucination Risk Estimation and Shift-Robust Selective Generation for Retrieval-Augmented Language Models

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Abstract

Retrieval-augmented generation (RAG) is widely deployed to reduce hallucination in large language models, yet retrieval is itself a noise source: irrelevant, contradictory, or misleading evidence can induce hallucinations a model would not otherwise produce. We study selective generation for RAG, where a system answers only when its hallucination risk is statistically controlled, under the realistic condition that the retrieval-noise distribution shifts between calibration and deployment. Using a controlled two-regime testbed that separates parametric from retrieval-induced error, we establish three behavioural regularities: a single unopposed false document is absorbed at 85 to 100 percent across three generators of two architectures, even against facts a model knows (context dominance); contradictions at the first context position are more than twice as damaging as at the last (misinformation primacy); and under conflicting evidence all errors are confident assertions rather than abstentions (silent failure). We then show that a noise-conditional risk estimator using observable retrieval descriptors predicts hallucination far better than model-internal uncertainty, beating sampling-based semantic entropy by 0.31 to 0.40 AUROC across six instruction-tuned model families. Standard conformal risk control and its covariate-shift remedies all violate a 10 percent target in 200 of 200 resplits under noise shift; we introduce risk-binned Mondrian conformal risk control, prove its validity for bin-measurable shifts, and

show it eliminates violations at competitive answer rates, with a label-free deployment monitor that makes its assumption checkable. Results hold across eight generators from 80M to 9B parameters, five model families, two architectures, and two content domains. Code and data are released.

Keywords: retrieval-augmented generation, hallucination, selective generation, conformal risk control, distribution shift, uncertainty quantification

1 Introduction

Retrieval-augmented generation (Lewis et al. 2020) grounds large language models (LLMs) in external corpora and is routinely presented as a mitigation for hallucination. The argument rests on an assumption that deployment rarely satisfies: that retrieved evidence is relevant and correct. Production retrieval pipelines return passages that are off-topic, partially relevant, stale, about confusable entities, or factually wrong (Chen et al. 2024; Wang et al. 2025; Cuconasu et al. 2024). Worse, the knowledge corpus itself can be corrupted: Zou et al. (2025) show that injecting a handful of crafted documents reliably steers RAG answers, and defenses remain weak. A growing literature documents that LLMs over-trust retrieved context even against their own parametric knowledge (Xie et al. 2024; Kortukov et al. 2024; Xu et al. 2024), so that RAG can paradoxically suppress appropriate abstention because added context raises confidence even when that context is wrong. When evidence is bad, RAG does not merely fail to help; it manufactures fluent, confident errors.

For high-stakes applications the question is therefore not only how to make RAG more robust, but when a RAG system should decline to answer. Selective prediction has a long history in classification (Chow 1970; Geifman and El-Yaniv 2017); for generation, conformal methods provide finite-sample guarantees on error rates among answered queries (Angelopoulos et al. 2024; Abbasi-Yadkori et al. 2024; Quach et al. 2024; Mohri and Hashimoto 2024). Existing conformal treatments of RAG, however, either certify population-level risk (Kang et al. 2024) or assume the calibration data is exchangeable with deployment (Li et al. 2024; Feng et al. 2025). Retrieval noise breaks exchangeability in a specific and practically common way: the mixture of noise regimes at deployment (which retriever, which corpus, which adversary) differs from the mixture at calibration. A system calibrated on clean development traffic and deployed against a degraded or poisoned corpus faces exactly this shift.

This paper studies per-decision risk control for RAG under retrieval-noise shift. We ask three questions. First, can hallucination be predicted before answering from observable signals, and how much do retrieval-side descriptors add over the model-internal uncertainty measures that dominate current practice (Kuhn et al. 2023; Farquhar et al. 2024; Kossen et al. 2024)? Second, how do noise type, severity, and position causally shape hallucination and abstention, and how do these effects interact with parametric knowledge? Third, when standard conformal calibration fails under noise shift, what restores validity, and at what cost in answer rate?

Causal answers to the second question require intervening on evidence while holding queries fixed, and require separating parametric error (the model would have been wrong anyway) from retrieval-induced error (the noise caused the mistake). Naturalistic benchmarks support neither. We therefore construct a controlled two-regime testbed. A synthetic world of 240 fictional scientists guarantees that no pretrained model holds parametric knowledge of the queried facts, so closedbook hallucination is 100% by construction and every correct answer is attributable to retrieval. Sixty well-known real facts (country capitals and currencies) provide the complementary regime in which mid-size models answer correctly without context 60% to 90% of the time. Typed noise is injected into a BM25 pipeline at graded severities and controlled context positions: partial and total irrelevance, contradictions at first and last rank, unopposed misinformation, confusable-entity distractors, and compound mixtures. Every generation is logged with retrieval-side, model-side, and closedbook-contrast features and labeled exactly against the closed answer space. The released dataset comprises 6,230 generations on the main domain across three generators spanning encoder-decoder and decoder-only architectures, plus 2,970 generations on an independent second domain.

Our contributions are as follows.

1. **Behavioural regularities of noisy RAG** (§5.1). We document three phenomena. Context dominance: one unopposed false document yields hallucination rates of 85% to 100% across all three generators, including on facts the models answer correctly without context. Misinformation primacy: a contradiction at rank 1 is more than twice as damaging as the same document at rank 4 (53% versus 23% hallucination without parametric support; 30% versus 0% with it). Universal silent failure: under every conflict-noise type, 100% of errors are confident assertions; behavioural abstention occurs only under irrelevance noise. The negative-rejection ability targeted by current instruction tuning (Chen et al. 2024) detects absent evidence but not deceptive evidence.
2. **Noise-conditional risk estimation** (§5.2). Combining retrieval descriptors, model-internal signals, and closedbook-contrast features raises hallucination-prediction AUROC from 0.788 to 0.915, with the margin over model-internal signals widening as noise becomes subtler and concentrated in conflict noise (AUROC 0.60 to 0.72), the regime in which models never warn.
3. **Shift-robust selective generation** (§5.4). Under clean-heavy calibration and noise-heavy deployment, marginal conformal risk control and its standard covariate shift remedies, learned-group calibration and density-ratio weighting (Tibshirani et al. 2019), violate a 10% risk target in every one of 200 resplits. Risk-binned Mondrian conformal risk control, which certifies score bins with Clopper-Pearson upper bounds, is proved valid for bin-measurable shifts (Proposition 1), eliminates violations entirely, answers 2.3 times more queries than a worst-group bound, and remains valid across target levels $\alpha \in \{0.10, 0.15, 0.20\}$, bin counts $B \geq 10$, and all tested shift severities.
4. **Transfer analysis** (§5.8). Risk estimators transfer across eight generators from 80M to 9B parameters and five model families with AUROC 0.87 to 0.96, yet transplanted thresholds realize 1.7 to 2.7 times the target risk. Discrimination is

substantially model-general; calibration is not. The binned certification scheme supports cheap local recalibration.

2 Related Work

2.1 Noise, conflict, and corruption in RAG

Benchmarks and analyses have established that retrieval quality governs RAG reliability. RGB isolates noise robustness and negative rejection as core abilities and finds models lacking in both (Chen et al. 2024). RAGTruth provides word-level hallucination annotations for RAG outputs (Wu et al. 2024). RAAT types retrieval noise and trains adapters against it (Fang et al. 2024); Astute RAG consolidates internal and external knowledge at inference (Wang et al. 2025). Cuconasu et al. (2024) report the counter-intuitive finding that random irrelevant documents can improve RAG accuracy, while related distractors hurt; our graded-irrelevance results refine this picture (§6). On the adversarial side, PoisonedRAG demonstrates effective knowledge corruption attacks with few injected documents (Zou et al. 2025). Position effects are documented for long contexts by Liu et al. (2024); we show the analogous effect for misinformation specifically. Knowledge-conflict studies characterise when models prefer context over memory (Xie et al. 2024; Kortukov et al. 2024; Xu et al. 2024). This literature measures and manipulates behaviour; it does not predict per-query hallucination risk or provide decision-level guarantees.

2.2 Uncertainty quantification for LLMs

Semantic entropy estimates uncertainty over meanings rather than strings (Kuhn et al. 2023; Farquhar et al. 2024); probes recover it cheaply from hidden states (Kossen et al. 2024). Self-evaluation signals include $P(\text{True})$ (Kadavath et al. 2022). These methods treat the generator as a closed system. None conditions on the quality or structure of retrieved evidence, and our results quantify the cost of that omission under retrieval noise.

2.3 Selective prediction and conformal guarantees

Selective classification with a reject option was introduced for deep networks by Geifman and El-Yaniv (2017). Conformal risk control bounds the expectation of monotone losses (Angelopoulos et al. 2024); conformal abstention applies this to hallucination (Abbasi-Yadkori et al. 2024). For RAG, TRAQ ensembles conformal sets over retrieval and generation (Li et al. 2024); Conformal-RAG filters sub-claims with group-conditional coverage (Feng et al. 2025); C-RAG certifies an upper bound on population-level generation risk, including under bounded distribution drift (Kang et al. 2024). Weighted conformal prediction corrects for known covariate shift via likelihood ratios (Tibshirani et al. 2019), adaptive conformal inference tracks unknown drift online (Gibbs and Candès 2021), and the limits of distribution-free conditional inference are themselves characterised by Barber et al. (2021). The guarantees in this literature are of three kinds, and none is the one a selective-answering system needs: conformal risk control and C-RAG bound population or marginal risk (Angelopoulos

et al. 2024; Kang et al. 2024); conditional conformal methods and Conformal-RAG bound conditional *coverage* rather than risk (Gibbs et al. 2025; Feng et al. 2025); and covariate-shift methods, including weighted conformal risk control, restore only *marginal* guarantees under a shifted test distribution (Tibshirani et al. 2019; Gibbs and Candès 2021; Zecchin and Simeone 2025). Since exact feature-conditional coverage is impossible distribution-free (Barber et al. 2021), we instead partition the feature space into a small number of risk bins. We show empirically that likelihood-ratio weighting and unsupervised grouping both fail under noise-regime selection, and that population certificates and per-decision validity diverge sharply here; our risk-binned Mondrian conformal risk control is, to our knowledge, the only method delivering per-query, risk-bin-conditional control of the answer-or-abstain risk that stays valid under measurable retrieval-noise covariate shift, needing neither ratio estimation nor exchangeability, only that the calibrated score resolves the shift.

3 Problem Setup and Method

3.1 Setting and observable features

Let q denote a query, $D = \{d_1, \dots, d_k\}$ the retrieved context, $a = f_\theta(q, D)$ the generated answer, and $y \in \{0, 1\}$ the indicator that a is a hallucination, defined as a confident assertion whose normalised form contradicts the gold answer. Behavioural abstentions (the model answers “unknown” or equivalent) form a separate outcome class and are analysed separately. Each example carries a feature vector $x = (x^{\text{ret}}, x^{\text{mod}}, x^\Delta)$.

Retrieval-side descriptors x^{ret} : mean, minimum, and top-1 BM25 relevance of the context; the top-1 to top-2 margin; score dispersion; an inter-document conflict flag computed by extracting values for the queried attribute from each context document and testing for disagreement; the number of extracted claims; and a context-presence indicator. All are computable at decision time without labels.

Model-side signals x^{mod} : mean and minimum token log-probability of the generated answer, mean, maximum, and first-token predictive entropy, and answer length, in the spirit of standard white-box uncertainty measures.

Closedbook-contrast features x^Δ : the change in mean answer log-probability and entropy relative to the same query answered with no context, and an indicator of whether the answer changed. These operationalise the confidence-gain idea of Bi et al. (2025) as risk features. They require one additional closedbook pass per query, which is cheap relative to retrieval.

3.2 Risk estimation

We estimate $\hat{r}(x) \approx \Pr[y = 1 \mid x]$ with gradient-boosted trees (150 trees, depth 3), evaluated with question-grouped five-fold cross-validation so that no question appears in both training and test folds. Feature-family ablations isolate the contribution of each signal class.

3.3 Selective generation and risk-binned certification

A selection rule answers query x iff $\hat{r}(x) \leq \tau$. Conformal risk control (CRC) selects τ on calibration data so that the hallucination rate among answered queries is at most a target α , with finite-sample correction. The guarantee requires calibration and deployment to be exchangeable. Retrieval-noise shift, modelled as a change in the mixture of noise regimes, violates this.

We propose risk-binned Mondrian CRC. Partition the calibration score range into B quantile bins $\{I_b\}_{b=1}^B$. For each bin compute the Clopper-Pearson upper confidence bound $\hat{p}_b^+$ at level $1 - \delta$ on the within-bin hallucination rate, and certify bin b iff $\hat{p}_b^+ \leq \alpha$. At deployment, answer iff the score falls in a certified bin. The scheme never extrapolates a threshold into score regions whose risk it has not measured; it only reuses bins whose risk it has certified.

Assumption 1 (Bin-measurable covariate shift) *The deployment distribution Q and calibration distribution P satisfy (i) $Q(y | x) = P(y | x)$, and (ii) the likelihood ratio $dQ/dP(x)$ is constant on each bin I_b .*

Condition (ii) formalises the requirement that the score resolves the shift: deployment may reweight bins arbitrarily, including adversarially toward high-risk bins, but may not redistribute mass within a bin. Selection over noise regimes that the risk score separates is of exactly this form.

Proposition 1 (Validity under bin-measurable shift) *Under Assumption 1, with probability at least $1 - B\delta$ over the calibration draw, the hallucination rate among answered deployment queries is at most α , simultaneously for every deployment distribution Q satisfying the assumption.*

Proof Fix a bin I_b . By Assumption 1(ii) the conditional distribution of x given $x \in I_b$ coincides under P and Q ; with (i), the within-bin risk $p_b = \Pr[y = 1 | x \in I_b]$ is distribution-independent. Calibration labels within I_b are i.i.d. Bernoulli(p_b), so $\Pr[p_b \leq \hat{p}_b^+] \geq 1 - \delta$ by the Clopper-Pearson construction. A union bound over the B bins gives, with probability at least $1 - B\delta$, $p_b \leq \alpha$ for every certified bin. The deployment risk among answered queries is a Q -weighted convex combination of certified-bin risks and is therefore at most α for any bin reweighting, which covers every Q in the assumption class simultaneously. $\square$

Remark 1 The guarantee requires no estimate of the likelihood ratio. This is the step at which weighted conformal methods fail in our experiments: when noise regimes overlap in feature space, the estimated ratio under-corrects precisely in the high-risk tail. The price of certification is conservativeness, quantified empirically in §5.4 and §5.5: within-bin score ordering is discarded, and small bins certify only with generous margins. The union-bound factor $B\delta$ ties the bin count to the confidence level; we use $B = 10$ and $\delta = 0.10$ in the main experiments and sweep both in §5.5.

Algorithm 1: Risk-binned Mondrian CRC for selective RAG**Input:** calibration set $\{(x_i, y_i)\}_{i=1}^n$, risk scorer $\hat{r}$, target α , bins B , level δ .

1. Compute scores $s_i = \hat{r}(x_i)$ and bin edges $\{q_b\}_{b=0}^B$ as $B+1$ empirical quantiles of $\{s_i\}$.
2. For each bin $b = 1, \dots, B$: let $S_b = \{i : q_{b-1} < s_i \leq q_b\}$, $k_b = \sum_{i \in S_b} y_i$, $n_b = |S_b|$; compute the Clopper-Pearson upper bound $\hat{p}_b^+ = \text{BetaInv}(1 - \delta, k_b + 1, n_b - k_b)$; certify bin b if $\hat{p}_b^+ \leq \alpha$.
3. At deployment, for query x : compute $s = \hat{r}(x)$, locate its bin $b(s)$; answer if $b(s)$ is certified, otherwise abstain.

Fig. 1 Risk-binned Mondrian conformal risk control.

We next characterise the price of certification in answer rate, which explains when the method is useful and how to choose the bin count.

Proposition 2 (Answer rate and its decomposition) *Let $\pi_b = \Pr_Q[\hat{r}(x) \in I_b]$ be the deployment mass of bin I_b and let $C \subseteq \{1, \dots, B\}$ index the certified bins. The answer rate of risk-binned Mondrian CRC is $\rho = \sum_{b \in C} \pi_b$, and a bin is certified whenever its empirical risk satisfies $\hat{p}_b \leq \alpha - \varepsilon_b(\delta, n_b)$, where $\varepsilon_b(\delta, n_b)$ is the Clopper-Pearson half-width at calibration count n_b . Consequently*

$$\rho \geq \sum_{b: p_b \leq \alpha - 2\varepsilon_b} \pi_b,$$

so the achievable answer rate is governed by the deployment mass lying in score bins whose true risk is bounded away from α by twice the finite-sample margin.

Proof The answer-rate identity is immediate since answered queries are exactly those whose score lands in a certified bin. For the bound, the empirical within-bin risk concentrates around p_b within $\varepsilon_b(\delta, n_b)$ with probability $1 - \delta$ by the Clopper-Pearson interval; whenever $p_b \leq \alpha - 2\varepsilon_b$, the upper bound $\hat{p}_b^+ = \hat{p}_b + \varepsilon_b \leq p_b + 2\varepsilon_b \leq \alpha$, so the bin is certified. Summing the deployment mass of these bins lower-bounds ρ . $\square$

Proposition 2 yields two practical consequences. First, the method answers nothing only when essentially all deployment mass sits in bins whose true risk exceeds α , that is, when the deployment stream is genuinely almost all high-risk; in that regime abstention is the correct behaviour. Second, because $\varepsilon_b = O(\sqrt{p_b(1-p_b)/n_b})$, increasing the bin count B shrinks each n_b and widens ε_b , which lowers answer rate, while decreasing B coarsens the partition and risks violating Assumption 1(ii). The two effects bracket an interior optimum, consistent with the bin-count sweep in §5.5, where $B = 10$ maximises answer rate subject to validity.

4 Experimental Design

4.1 Two-regime fact world

The corpus contains 2,200 documents. For each of 240 fictional scientists we generate five gold fact sentences (birth year, birth city, field, invention, institution) from templates with filler clauses; for each of 60 real facts (30 capitals, 30 currencies) we write one gold sentence. For every question we add one confident contradiction document asserting a plausible false value, framed as a revised archival review or administrative reform; for every synthetic question, one confusable-entity distractor whose subject differs from the queried entity by one character; and 400 irrelevant topical fillers. Synthetic entities make parametric knowledge impossible, so closedbook hallucination is 100% by construction; real facts provide the regime where parametric knowledge exists (closedbook accuracy: flan-t5-base 60%, Qwen2.5-0.5B-Instruct 90%).

4.2 Conditions

Eleven conditions per question. Clean retrieval; graded irrelevance irr25, irr50, irr75, irr100 (one to four of the $k = 4$ context slots replaced by topically plausible irrelevant documents, with the gold document removed at 100%); contra_r1 and contra_r4 (gold present, contradiction at first or last rank); contra_only (contradiction present, gold absent); distractor; mixed (gold plus contradiction plus irrelevant); closedbook.

4.3 Models, pipeline, and labels

BM25 retrieval over the full corpus; greedy decoding with at most 12 to 14 new tokens. Generators: flan-t5-base (250M, encoder-decoder, all 11 conditions, 3,300 generations), flan-t5-small (80M, 6 conditions, 1,800 generations), and Qwen2.5-0.5B-Instruct (decoder-only, chat-templated prompts, 4 conditions on a 100-question subset, 410 generations), plus a 720-generation preliminary campaign used only for the noise-subtlety comparison. Answers are normalised and matched exactly against the closed answer space, with a pattern-based abstention class. Labels are therefore programmatic and exact, which removes LLM-judge circularity from the measurement loop. The harness checkpoints atomically, resumes from interruption, and runs on commodity CPU hardware; a companion notebook scales the identical protocol to 7B-class models on GPU.

4.4 Evaluation protocol

Risk estimation uses question-grouped five-fold cross-validation; paired bootstrap over questions (2,000 resamples) gives confidence intervals for AUROC differences. Conformal experiments use 200 question-level resplits; in each, calibration is drawn clean-heavy (noisy-condition rows subsampled to 30%) and deployment noise-heavy (clean rows subsampled to 50%), emulating calibration on development traffic followed by deployment against a degraded corpus. Sensitivity analyses vary the target α , the bin count B , and the calibration noise retention from 0.9 down to 0.1.

Table 1 Hallucination (H) and abstention (A) rates by condition and regime, flan-t5-base ($n = 240$ synthetic and 60 real per cell). C denotes correct.

Condition	Synthetic (no param. knowledge)		Real (param. knowledge)	
	H	A	H	A
clean	0.00	0.00	0.00	0.00
irr25	0.03	0.00	0.00	0.00
irr50	0.09	0.00	0.00	0.00
irr75	0.16	0.00	0.00	0.00
irr100	0.62	0.38	0.20	0.78
contra_r1	0.53	0.00	0.30	0.00
contra_r4	0.23	0.00	0.00	0.00
contra_only	1.00	0.00	1.00	0.00
distractor	0.17	0.00	0.00	0.00
mixed	0.45	0.00	0.15	0.00
closedbook	1.00	0.00	0.40	0.00

5 Results

5.1 Behavioural regularities of noisy RAG

Context dominance.

Under `contra_only`, hallucination is 100% in both regimes for flan-t5-base (Table 1). The decoder-only replication is equally stark: Qwen2.5-0.5B-Instruct, which answers the real-fact questions correctly 90% of the time with no context, hallucinates on 85% of them when a single unopposed false document is present and on 100% of the synthetic questions. Parametric knowledge arbitrates conflicts only when supporting evidence is also retrieved; in its absence, context wins almost absolutely. This sharpens the parametric-bias picture of Kortukov et al. (2024) and gives a security reading consistent with Zou et al. (2025): a corpus-poisoning attack need not outvote the truth, only keep it absent from the retrieved window, and we quantify the per-document absorption rate that makes such attacks efficient.

Misinformation primacy.

With gold evidence present, moving the contradiction from rank 1 to rank 4 cuts hallucination from 0.53 to 0.23 without parametric support and from 0.30 to 0.00 with it. Position in the context window is a deployment-observable risk factor, complementary to the attention position effects of Liu et al. (2024), and no existing uncertainty or selective-RAG method conditions on it.

Universal silent failure.

Across every conflict condition (`contra_r1`, `contra_r4`, `contra_only`, `distractor`, `mixed`), 100% of errors are confident assertions; the model abstained in 0 of those error cases. Under pure irrelevance the model abstains on 38% (synthetic) to 78% (real) of affected queries. The same asymmetry holds exactly for the decoder-only model. Training for

Table 2 Hallucination-risk prediction, question-grouped five-fold CV, flan-t5-base ($n = 3,161$ answered; base rate 0.363).

Feature family	AUROC
Retrieval-side descriptors	0.897
Model-internal (logprob, entropy)	0.788
Closedbook contrast	0.700
Model + contrast	0.811
Combined (noise-conditional)	0.915

negative rejection (Chen et al. 2024) therefore addresses the loud failure mode and leaves the silent one untouched, which is the central argument for external, evidence-aware risk control.

Chat tuning increases conflict vulnerability.

With gold and contradiction both at rank 1, the chat-tuned decoder hallucinates on 73% of real-fact queries against 30% for flan-t5-base, despite higher closedbook accuracy (90% versus 60%); stronger context obedience, an explicit objective of RAG instruction tuning, appears to trade against conflict robustness, a trade-off worth testing at larger scales with the released harness.

5.2 Noise-conditional risk estimation

The combined estimator improves on model-internal signals by 0.127 AUROC (95% CI 0.115 to 0.140); the closedbook-contrast family alone adds 0.023 (CI 0.013 to 0.035) over model-internal signals (Table 2). The comparison with our preliminary campaign is instructive. With crude noise (only total irrelevance and rank-1 contradictions), the combined-versus-model gap was 0.026; with the graded and positional noise grid it is 0.127, because model-internal discrimination falls from 0.905 to 0.788 while retrieval descriptors hold (0.899 to 0.897). Subtle noise is precisely where context-agnostic uncertainty fails and noise conditioning pays.

Where prediction remains hard.

Per-condition diagnostics (out-of-fold scores) show near-perfect discrimination on irrelevance noise (irr25 to irr100: AUROC 0.98 to 1.00) and substantially lower discrimination under conflict (contra_r1: 0.605; mixed: 0.596; contra_r4: 0.722; distractor: 0.839). The estimator is well calibrated overall (ECE 0.024, 10 quantile bins), and feature importances are dominated by mean retrieval relevance (0.45) followed by minimum token log-probability and the relevance margin. The hard residual case, conflict-induced hallucination, is exactly the silent regime of §5.1; this conjunction, hardest to predict and never self-signalled, is what the conformal layer must cover.

Table 3 Hallucination prediction against real UQ baselines computed on the same generator (flan-t5-small, 120-question subset, five samples per query for semantic entropy). Higher is better. The noise-conditional estimator uses only label-free retrieval descriptors.

Signal	AUROC
Semantic entropy (Kuhn et al. 2023; Farquhar et al. 2024)	0.675
$1 - P(\text{True})$ (Kadavath et al. 2022)	0.710
Noise-conditional estimator (ours, retrieval only)	0.893
Noise-conditional estimator + semantic entropy + $P(\text{True})$	0.915

5.3 Comparison with uncertainty-quantification baselines

A natural question is whether the gains over model-internal log-probability and entropy (§5.2) survive comparison with purpose-built uncertainty-quantification methods. We computed two standard baselines on the same generator: discrete semantic entropy (Kuhn et al. 2023; Farquhar et al. 2024) from five temperature samples, using exact normalised-string equivalence as the semantic-equivalence relation (which is exact for the closed answer space rather than approximated by an entailment model), and $P(\text{True})$ self-evaluation (Kadavath et al. 2022). Table 3 shows the noise-conditional estimator outperforms semantic entropy by 0.218 AUROC (95% CI 0.193 to 0.243) and $P(\text{True})$ by 0.183. Adding both signals to our estimator yields only a further 0.022 (95% CI 0.012 to 0.033), so the predictive power resides in the observable retrieval structure, not in sampling-based self-consistency. This is consistent with the mechanism we isolate in §5.10: sampling-based entropy tracks the model’s internal disagreement, which is informative when evidence is missing but near blind when the model confidently absorbs a coherent falsehood, so under retrieval noise the decisive signal is the quality of the evidence, which model-internal uncertainty does not observe. The gap widens with model capability: on Qwen2.5-7B-Instruct, sampling-based semantic entropy is barely better than chance at predicting retrieval-induced hallucination (AUROC 0.536), while the noise-conditional estimator reaches 0.892, a difference of 0.356 AUROC (95% CI 0.340 to 0.373) – larger than the 0.218 gap on the 80M model, the opposite of what a small-model artefact would predict.

5.4 Selective generation under noise shift

Table 4 contains the central result. Marginal CRC roughly doubles its target risk. The two textbook covariate-shift remedies fail in every resplit as well: unsupervised k -means groups in descriptor space do not align with risk strata, and the logistic-regression density ratio under-corrects in the high-risk tail because noise regimes overlap in feature space. The worst-group bound is valid in most resplits but answers only 12.7% of queries. Risk-binned certification eliminates violations entirely at a 29.2% answer rate. The mechanism is the one identified in §3.3: certification reuses only measured risk, so reweighting bins cannot break it, whereas every thresholding method implicitly extrapolates.

Table 4 Selective generation at target risk $\alpha = 0.10$ under clean-heavy calibration and noise-heavy deployment; 200 resplits, flan-t5-base. Violation is the fraction of resplits whose realized risk exceeds α .

Method	Realized risk	Violation	Answer rate
Marginal CRC	0.215	100%	70.5%
Heuristic-group CRC	0.189	100%	60.9%
Learned-group CRC (k -means)	0.213	100%	67.0%
Density-ratio weighted CRC	0.176	100%	61.5%
Worst-group CRC	0.035	22%	12.7%
Risk-binned Mondrian CRC (ours)	0.063	0%	29.2%

5.5 Sensitivity and robustness analyses

Three sweeps (60 resplits per cell; full figures in the repository) confirm the guarantee is not an artefact of one operating point. Across targets, risk-binned certification is valid at $\alpha \in \{0.10, 0.15, 0.20\}$ with zero violations and shows 8% violations at the strict $\alpha = 0.05$ (consistent with the certification level $\delta = 0.10$), while marginal CRC violates at every target with realized risk roughly 2α . Across bin counts, $B = 5$ is too coarse (25% violations) while $B \in \{10, 20, 40\}$ are all valid, trading answer rate (0.29 down to 0.20) against granularity exactly as the union-bound analysis predicts. Across shift severities, marginal risk grows monotonically from 0.143 to 0.291 as calibration sees less noise, while risk-binned certification stays controlled throughout.

5.6 Robustness to within-bin shift and a deployment monitor

Proposition 1 assumes the shift is bin-measurable (Assumption 1(ii)); its one failure mode is a within-bin shift that raises a certified bin’s true risk without moving scores. Holding the inter-document conflict flag out of the risk score so that it acts as a latent within-bin risk driver, we construct an adversarial deployment that upweights high-conflict items twelvefold. Even then the method degrades only mildly: realized risk rises from 0.066 to 0.072 and the violation rate from 0% to 4% against the 0.10 target, because the score already absorbs most of the risk-relevant covariate structure. When stricter safety is required, the held-out covariate can be monitored at deployment without labels: decertifying a bin when its deployment conflict rate exceeds calibration flags 4.67 bins on average under the adversarial within-bin shift but only 1.39 under benign reweighting (a 3.4-fold contrast), making the method’s one assumption checkable in deployment rather than merely assumed.

5.7 Generality across models, estimators, and domains

Six instruction-tuned families.

To rule out that the result is an artifact of one model or one architecture, we ran the full eleven-condition grid on six decoder-only instruction-tuned generators spanning 1.5B to 9B parameters and four independent pretraining families: Qwen2.5 (1.5B, 3B, 7B), Microsoft Phi-3.5-mini (3.8B), Mistral-7B-v0.3, and Google Gemma-2-9B. Each run

Table 5 Generality of the central result across eight generators spanning 80M to 9B parameters and five model families (FLAN-T5, Qwen, Microsoft Phi, Mistral, Google Gemma), two architectures (encoder-decoder and decoder-only), and an independent second domain. For every generator the combined risk estimator discriminates hallucination well (AUROC 0.87 to 0.96), the sampling-based semantic entropy baseline stays near chance (AUROC 0.52 to 0.64), marginal CRC violates the $\alpha = 0.10$ target in essentially every resplit, and risk-binned Mondrian CRC restores the guarantee (0% to 4% violations). The six decoder-only instruction-tuned models each ran the full eleven-condition grid (about 3,300 generations) with white-box logprob and entropy features and bidirectional-NLI semantic entropy. AUROC columns use question-grouped cross-validation.

Generator	Domain	Estimator AUROC		Marg.	Risk-binned CRC	
		Comb.	SE	Viol.	Viol.	Ans. rate
flan-t5-small (80M, enc-dec)	scientists	0.959	—	100%	0%	17.5%
flan-t5-base (250M, enc-dec)	scientists	0.915	—	100%	0%	28.8%
Qwen2.5-1.5B (dec-only)	scientists	0.871	0.535	100%	0%	35.3%
Qwen2.5-3B (dec-only)	scientists	0.874	0.530	100%	4%	29.7%
Phi-3.5-mini (3.8B, dec-only)	scientists	0.948	0.640	100%	0%	28.9%
Mistral-7B-v0.3 (dec-only)	scientists	0.950	0.617	100%	0%	36.9%
Qwen2.5-7B-Instruct (dec-only)	scientists	0.892	0.536	100%	1%	36.7%
Gemma-2-9B-it (dec-only)	scientists	0.923	0.523	100%	1%	39.0%
flan-t5-small (80M, enc-dec)	companies	0.920	—	100%	0%	21.7%

logs true token logprobs, predictive entropy, and bidirectional-NLI semantic entropy, for about 3,300 generations per model. The combined noise-conditional estimator discriminates hallucination with AUROC between 0.871 and 0.950 on every family, while sampling-based semantic entropy never rises above 0.640 and sits near chance (0.52 to 0.54) on four of the six. The combined-minus-semantic-entropy advantage ranges from +0.308 to +0.400 AUROC, and every per-model bootstrap interval excludes zero by a wide margin, so the baseline gap is not specific to any one generator. Marginal CRC violates the $\alpha = 0.10$ target in 100% of resplits on all six models, and risk-binned Mondrian CRC restores the guarantee on all six (realized risk 0.022 to 0.060, violation 0% to 4%) at answer rates of 29% to 39%. The largest semantic-entropy gap (+0.400) and the highest answer rate (39%) both occur on the largest model, Gemma-2-9B, so neither the estimator advantage nor the efficiency of the guarantee degrades with scale.

Second domain.

On an independent domain with disjoint subject matter (240 fictional technology companies and 30 real chemical-element facts, 2,970 generations with flan-t5-small), every phenomenon reproduces: unopposed misinformation is absorbed at 100% (synthetic) and 73% (real), the rank-1 contradiction is 0.72 against 0.18 at rank 4, and silent failure is near-universal. The estimator reaches AUROC 0.920, marginal CRC violates the target in every resplit, and risk-binned CRC never violates it (realized risk 0.016). The regularities and the guarantee are properties of noisy RAG, not of one corpus.

Scaling and the position effect.

At 7B (Qwen2.5-7B-Instruct), negative rejection improves with scale, so silent failure under pure irrelevance falls to zero, yet under unopposed misinformation 97% of errors remain confident assertions. The position effect inverts relative to the encoder-decoder models: a contradiction at the last context position is more damaging than at the first (rank-1 0.38 versus rank-4 0.67), so position remains a major observable risk factor whose damaging end is architecture-dependent.

Estimator architecture.

The signal is not tied to gradient boosting: on flan-t5-base, logistic regression, random forest, boosting, and a small multilayer perceptron reach AUROC 0.870, 0.922, 0.915, and 0.905, so the gain comes from the features rather than the model class.

Recalibration.

Isotonic recalibration of the already well-calibrated score (ECE 0.024) does not improve the answer rate, so we report the method without it; on a poorly calibrated base scorer it would be expected to help.

5.8 Cross-model transfer

Training the risk estimator on one generator and deploying it on the other preserves discrimination but not calibration. From flan-t5-small to flan-t5-base, AUROC is 0.827 and a source-calibrated threshold at $\alpha = 0.10$ realizes risk 0.170; in the reverse direction, AUROC is 0.908 and realized risk is 0.269. Rankings of risky inputs are substantially model-general; absolute risk levels are not. Portable selective RAG therefore requires local recalibration, which binned certification supports at low cost since bins can be re-certified on a small labelled sample from the target system. We register the prediction that the same dissociation governs cross-lingual transfer, for instance from English to Bengali retrieval corpora, and leave its test to follow-up work on naturalistic multilingual data.

5.9 Naturalistic validation on real retrieval

The controlled world isolates each noise mechanism, but a referee may reasonably ask whether the effects survive genuine retrieval. We therefore rebuilt the pipeline on SQuAD: a 300-paragraph pool with one question per paragraph, real BM25 and dense (e5 plus FAISS) retrieval, the same typed-noise grid, and PoisonedRAG poisoning (Zou et al. 2025) as an adversarial condition, run on two instruction-tuned generators (Qwen2.5-7B and Mistral-7B-v0.3) with white-box features and discrete semantic entropy. Table 6 reports the estimator on the answered subset. The combined estimator reaches AUROC 0.84 to 0.86 on genuine retrieval for both generators, and BM25 and dense retrieval give almost identical numbers (0.842 versus 0.840 for Qwen, 0.852 versus 0.857 for Mistral), so the signal is retriever-independent. The advantage over semantic entropy is positive and significant in every cell (+0.14 to +0.31 AUROC, all confidence intervals excluding zero), though smaller than in the controlled world because Mistral’s semantic entropy is itself more informative here (0.71).

Table 6 Naturalistic validation on real SQuAD retrieval. Combined risk estimator versus the semantic entropy (SE) baseline on the answered subset, for two generators and two retrievers. Combined-minus-SE shows a bootstrap 95% confidence interval.

Generator	Retriever	Combined	SE	Combined – SE [95% CI]
Qwen2.5-7B	BM25	0.842	0.533	+0.309 [+0.284, +0.334]
Qwen2.5-7B	dense	0.840	0.545	+0.295 [+0.267, +0.325]
Mistral-7B	BM25	0.852	0.714	+0.138 [+0.117, +0.159]
Mistral-7B	dense	0.857	0.693	+0.164 [+0.141, +0.186]

Table 7 Decomposition of naturalistic injected failures into missing-evidence and misinformation, with each signal’s AUROC for predicting hallucination. Semantic entropy (SE) is informative on missing-evidence but near chance on misinformation; retrieval descriptors are the mirror image; the combined estimator captures both.

Generator	Failure mode	n	H-rate	SE	Retrieval	Combined
Qwen2.5-7B	missing-evidence	51	0.69	0.668	0.513	0.420
Qwen2.5-7B	misinformation	961	0.39	0.521	0.800	0.859
Mistral-7B	missing-evidence	453	0.86	0.701	0.504	0.744
Mistral-7B	misinformation	692	0.30	0.587	0.667	0.755

Per-condition hallucination on real retrieval confirms the controlled ordering: PoisonedRAG poisoning is the strongest inducer (0.74 to 0.85 across models), while the clean condition already hallucinates at 0.15 to 0.28. Because the clean base rate is this high, a 0.10 selective-risk target sits below it for these models, so the realistic operating point is $\alpha = 0.20$. At that target the dissociation reproduces directionally: marginal CRC violates between 47% and 62% of resplits, whereas risk-binned Mondrian CRC holds (0% to 1% violations, realized risk 0.09 to 0.12) at a 21% to 23% answer rate. We frame this honestly. The controlled world produces the dramatic headline that marginal CRC violates in essentially every resplit; naturalistic noise is milder and the base rates higher, so marginal CRC violates about half the resplits while the binned fix restores validity at a usable answer rate. The naturalistic run corroborates the direction of the controlled result on real data rather than reproducing its magnitude.

5.10 Why semantic entropy fails: missing-evidence versus misinformation

The baseline collapse and the naturalistic milder gap have a single explanation, which also reconciles the controlled and natural results. Within the injected naturalistic conditions we split failures into two mechanisms and measure each signal’s AUROC for predicting hallucination (Table 7). *Missing-evidence* failures (closed-book and graded irrelevance) leave the model with no usable passage; *misinformation* failures (rank-1 contradiction, contradiction-only, poisoning, and mixed) place a coherent but false passage in context.

The pattern is clean and complementary. Semantic entropy detects missing-evidence uncertainty (AUROC 0.67 to 0.70) but is near chance on confident misinformation absorption (0.52 to 0.59): when the model absorbs a false but coherent passage, its sampled answers agree on the absorbed falsehood, so entropy is low even though the answer is wrong; this is precisely the regime in which sampling-based entropy is known to be ineffective (Ma et al. 2025), and it aligns with the epistemic-versus-aleatoric view of language-model uncertainty (Ahdritz et al. 2024). The retrieval descriptors are the mirror image, near chance on missing-evidence (0.50 to 0.51, because retrieval scores do not reveal the model’s internal state) but strong on misinformation (0.67 to 0.80, because contradictory and irrelevant passages have observable signatures). The combined estimator captures both modes (Table 7). This reconciles the whole paper. It explains why the semantic entropy baseline collapses on injected conflict noise, the headline of §5.3 and §5.7; why semantic entropy looked competitive on purely natural noise (§5.11), which is dominated by missing-evidence failures; and why the combined estimator is the right design, since its two signal families are complementary rather than redundant. The Qwen missing-evidence cell has small $n = 51$ because that model abstains heavily under irrelevance, so Mistral ($n = 453$) is the reliable demonstration; the pattern is consistent across both generators.

5.11 Purely natural retrieval noise

To rule out the objection that injected noise is synthetic, we ran the pipeline with *no* injection at all: the full SQuAD pool of about 2,000 paragraphs, Qwen2.5-7B, BM25 and dense retrieval pooled, 553 answered queries at a base hallucination rate of 0.212. The retriever’s own failures suffice to induce hallucination. When the gold paragraph is retrieved, hallucination is 0.179 ($n = 507$); when the retriever misses the gold paragraph it rises to 0.625 ($n = 16$); and closed-book sits at 0.533 ($n = 30$). A natural retrieval miss therefore raises hallucination about 3.5 times with zero injected noise. On this subtle natural noise the retrieval descriptors are weak and semantic entropy is competitive (combined-minus-semantic-entropy +0.017, 95% CI $[-0.024, +0.059]$, not significant), the opposite of the injected finding. The decomposition of §5.10 explains why: natural retrieval noise is dominated by missing-evidence failures, the regime in which semantic entropy works and the retrieval descriptors do not.

6 Discussion

Reconciling with the power of noise.

Cuconasu et al. (2024) found that random unrelated documents can improve RAG accuracy whereas high-ranked distractors hurt. Our irrelevance fillers are topically plausible hard negatives, and their effect is monotonically harmful (0.00 to 0.62 hallucination as severity rises): harm scales with the plausibility of the noise, not its volume, and becomes silent exactly when plausibility is highest.

Per-decision versus population certificates.

C-RAG (Kang et al. 2024) certifies whether a RAG system is safe on average under bounded drift; we show a system can satisfy such a bound while per-decision calibration fails on every resplit under noise-regime selection. The two are complementary: population certificates for system acceptance, per-decision certification for runtime gating.

What the testbed buys and what it costs.

The controlled world provides exact labels, causal attribution of retrieval-induced error, and full coverage of a typed noise grid, none of which naturalistic benchmarks offer; the cost is ecological validity (template text, lexically detectable noise, BM25 retrieval). We regard it as a wind tunnel for RAG, and the phenomena established here are mechanism-level claims, replicated across two architectures, two content domains, six model families up to 9B, and naturalistic SQuAD retrieval with two retrievers.

Limitations.

Exact-match labels undercount paraphrase hallucinations, mitigated here by the closed answer space; on the naturalistic data we additionally validate labels against an independent large-model judge. Assumption 1(ii) is an idealisation; empirically the guarantee held across 200 adversarial resplits and all sensitivity sweeps, the within-bin stress test of §5.6 degraded it only to a 4% violation rate, and the deployment monitor detects the residual failure mode, but a shift that splits bins along an unmonitored covariate could still degrade validity. The certification answer rate (29% at $\alpha = 0.10$ under severe shift) is a substantial price; richer scores, adaptive binning, and partial within-bin ordering are natural directions. Our largest generator is 9B; the cross-architecture, cross-domain, and naturalistic agreement address, but do not close, the frontier-scale question.

A Reproducibility

All corpora are generated by released scripts from fixed seeds; the released dataset comprises 6,230 main-domain, 2,970 second-domain, and the naturalistic generations, all with deterministic exact-match labels and a pattern-based abstention class. Generators are public (flan-t5-base/small, Qwen2.5-1.5B/3B/7B, Mistral-7B-v0.3, Gemma-2-9B, Phi-3.5-mini). The risk estimator is gradient-boosted trees (150 trees, depth 3) under question-grouped five-fold cross-validation, robust across logistic regression, random forest, and a multilayer perceptron. The conformal protocol uses 200 question-level resplits with clean-heavy calibration (30% noisy retention) and noise-heavy deployment (50% clean retention), with $\alpha = 0.10$, $B = 10$, and $\delta = 0.10$ unless swept; AUROC differences use a 2,000-resample paired bootstrap over questions. Complete code, data, cached out-of-fold scores, and the analysis scripts that regenerate every table and figure are released at the repository linked in the Declarations.

Declarations.

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Author contributions.

Anmita Das and Shoumya Chowdhury contributed equally to this work. A.D. and S.C. led the conceptualisation and methodology, designed and implemented the experimental framework and analysis pipeline, ran the experiments, prepared the figures, and wrote the original draft. S.P. contributed to the conceptualisation and the framing of the selective-generation problem, advised on the experimental design and the statistical methodology, validated and interpreted the results, secured the computational resources, administered the project, and revised the manuscript through critical review and editing. All authors read and approved the final manuscript.

Use of AI tools.

The authors used AI tools only for code debugging and spell checking. All experimental designs, analyses, results, and claims are the authors' own and were verified by the authors, who take full accountability for the content of this article.

Data availability.

All datasets generated and analysed in this study, together with the complete code for corpus construction, noise injection, generation, and analysis, are publicly available at <https://github.com/shoumya-chy/ACML-RAG>. The study uses only publicly available, openly licensed models and synthetically generated corpora; no human-subjects or proprietary data are used. The naturalistic validation uses the publicly available SQuAD dataset together with the openly licensed Qwen2.5-7B-Instruct and Mistral-7B-v0.3 generators.

References

- Abbasi-Yadkori Y, Kuzborskij I, Stutz D, et al (2024) Mitigating LLM hallucinations via conformal abstention. arXiv preprint arXiv:240501563
- Ahdritz G, Qin T, Vyas N, et al (2024) Distinguishing the knowable from the unknowable with language models. In: International Conference on Machine Learning
- Angelopoulos AN, Bates S, Fisch A, et al (2024) Conformal risk control. International Conference on Learning Representations
- Barber RF, Candès EJ, Ramdas A, et al (2021) The limits of distribution-free conditional predictive inference. *Information and Inference: A Journal of the IMA* 10(2):455–482

- Bi B, Huang S, Wang Y, et al (2025) Parameters vs. context: Fine-grained control of knowledge reliance in language models. arXiv preprint arXiv:250315888
- Chen J, Lin H, Han X, et al (2024) Benchmarking large language models in retrieval-augmented generation. In: Proceedings of the AAAI Conference on Artificial Intelligence
- Chow CK (1970) On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16(1):41–46
- Cuconasu F, Trappolini G, Siciliano F, et al (2024) The power of noise: Redefining retrieval for RAG systems. In: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp 719–729
- Fang F, Bai Y, Ni S, et al (2024) Enhancing noise robustness of retrieval-augmented language models with adaptive adversarial training. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics
- Farquhar S, Kossen J, Kuhn L, et al (2024) Detecting hallucinations in large language models using semantic entropy. Nature 630:625–630
- Feng N, Sui Y, Hou Y, et al (2025) Response quality assessment for retrieval-augmented generation via conditional conformal factuality. In: Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval
- Geifman Y, El-Yaniv R (2017) Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems
- Gibbs I, Candès E (2021) Adaptive conformal inference under distribution shift. In: Advances in Neural Information Processing Systems
- Gibbs I, Cherian JJ, Candès EJ (2025) Conformal prediction with conditional guarantees. Journal of the Royal Statistical Society Series B: Statistical Methodology
- Kadavath S, Conerly T, Askell A, et al (2022) Language models (mostly) know what they know. arXiv preprint arXiv:220705221
- Kang M, Gürel NM, Yu N, et al (2024) C-RAG: Certified generation risks for retrieval-augmented language models. In: International Conference on Machine Learning
- Kortukov E, Rubinstein A, Nguyen E, et al (2024) Studying large language model behaviors under context-memory conflicts with real documents. In: First Conference on Language Modeling
- Kossen J, Han J, Razzak M, et al (2024) Semantic entropy probes: Robust and cheap hallucination detection in LLMs. arXiv preprint arXiv:240615927

- Kuhn L, Gal Y, Farquhar S (2023) Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In: International Conference on Learning Representations
- Lewis P, Perez E, Piktus A, et al (2020) Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems
- Li S, Park S, Lee I, et al (2024) TRAQ: Trustworthy retrieval augmented question answering via conformal prediction. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics
- Liu NF, Lin K, Hewitt J, et al (2024) Lost in the middle: How language models use long contexts. Transactions of the Association for Computational Linguistics 12:157–173
- Ma H, Pan J, Liu J, et al (2025) Semantic energy: Detecting LLM hallucination beyond entropy. arXiv preprint arXiv:250814496
- Mohri C, Hashimoto T (2024) Language models with conformal factuality guarantees. In: International Conference on Machine Learning
- Quach V, Fisch A, Schuster T, et al (2024) Conformal language modeling. In: International Conference on Learning Representations
- Tibshirani RJ, Barber RF, Candès E, et al (2019) Conformal prediction under covariate shift. Advances in Neural Information Processing Systems 32
- Wang F, Wan X, Sun R, et al (2025) Astute RAG: Overcoming imperfect retrieval augmentation and knowledge conflicts for large language models. In: Findings of the Association for Computational Linguistics
- Wu Y, Zhu J, Xu S, et al (2024) RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics
- Xie J, Zhang K, Chen J, et al (2024) Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In: International Conference on Learning Representations
- Xu R, Qi Z, Guo Z, et al (2024) Knowledge conflicts for LLMs: A survey. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing
- Zecchin M, Simeone O (2025) Generalization and informativeness of weighted conformal risk control under covariate shift. arXiv preprint arXiv:250111413
- Zou W, Geng R, Wang B, et al (2025) PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In: 34th USENIX Security Symposium, pp 3827–3844